CS-471: Machine Learning

Week 6

Instructor: Dr. Hashir Kiani

Classroom Etiquette and Participation Guidelines

- Treat everyone with respect and courtesy at all times
- Speak only when it is your turn and raise your hand before contributing to the discussion
- Do not interrupt others while they are speaking
- Join the class on time
- Keep mobile phones on silent during the lecture

Lecture Outline

- Recap of Previous Lecture
- Gini Impurity
- Entropy and Information Gain

Recap of Previous Lecture

- What is feature selection? Why is it required?
- What is feature engineering? Why is it required?
- What is regularization? Two types?
- Formulas for both types? Gradient with regularization?
- What is regularization parameter? What does it do?
- How is an optimum value of lambda determined?
- What is early stopping?
- What are decision trees used for?
- In what situations are decision trees more appropriate?
- How do we make a decision tree?

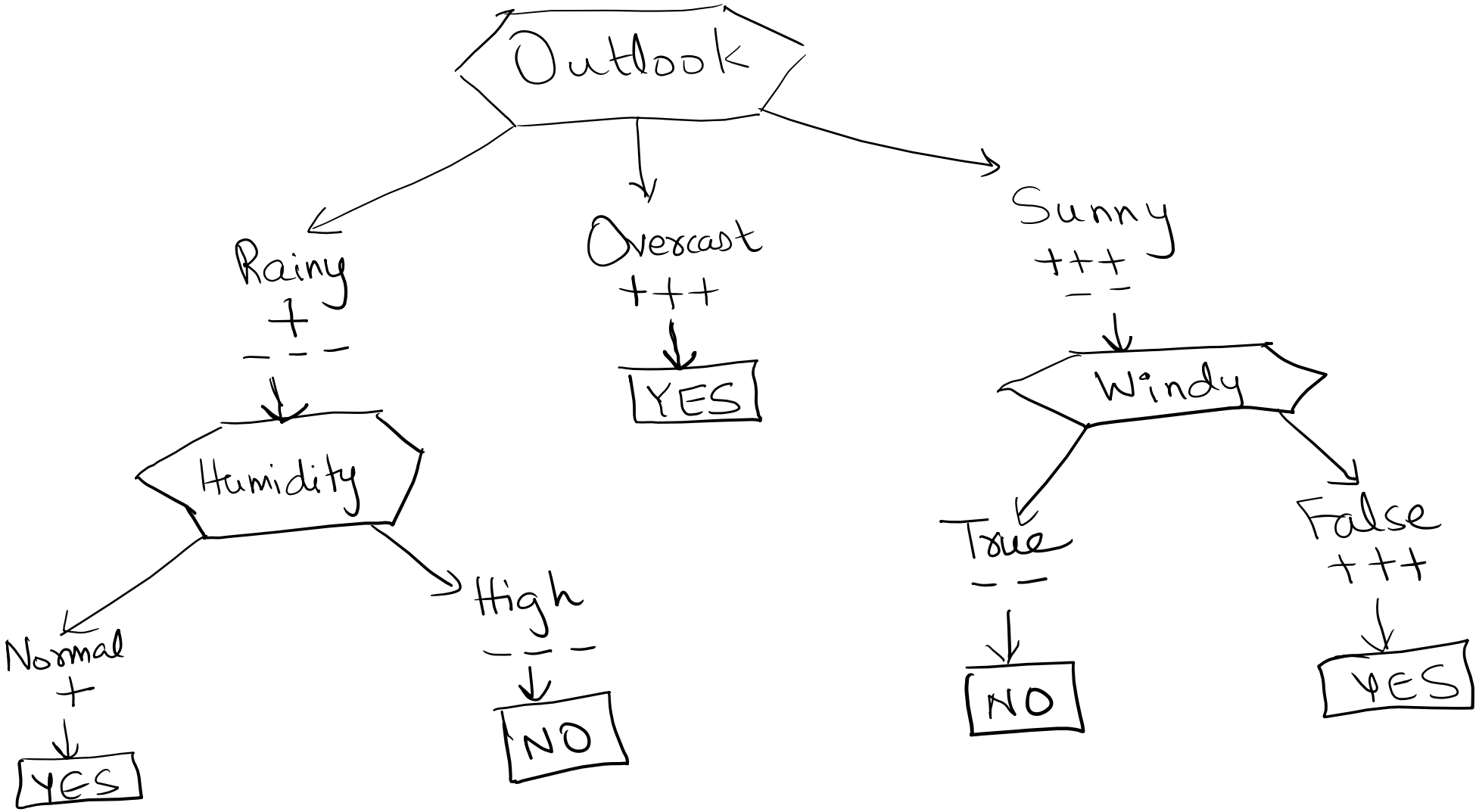
Training Set

Example	Input Attributes				Output Plays Golf
	Outlook	Temperature	Humidity	Windy	
$\boldsymbol{x}_1$	Rainy	Hot	High	False	$y_1 = \text{No}$
$\boldsymbol{x}_2$	Rainy	Hot	High	True	$y_2 = \text{No}$
$\boldsymbol{x}_3$	Overcast	Hot	High	False	$y_3 = \text{Yes}$
$\boldsymbol{x}_4$	Sunny	Mild	High	False	$y_4 = \text{Yes}$
$\boldsymbol{x}_5$	Sunny	Cool	Normal	False	$y_5 = \text{Yes}$
$\boldsymbol{x}_6$	Sunny	Cool	Normal	True	$y_6 = \text{No}$
$\boldsymbol{x}_7$	Overcast	Cool	Normal	True	$y_7 = \text{Yes}$
$\boldsymbol{x}_8$	Rainy	Mild	High	False	$y_8 = \text{No}$
$\boldsymbol{x}_9$	Sunny	Mild	Normal	False	$y_9 = \text{Yes}$
$\boldsymbol{x}_{10}$	Rainy	Mild	Normal	True	$y_{10} = \text{Yes}$
$\boldsymbol{x}_{11}$	Overcast	Mild	High	True	$y_{11} = \text{Yes}$
$\boldsymbol{x}_{12}$	Sunny	Mild	High	True	$y_{12} = \text{No}$

Validation Set

Example	Input Attributes				Output Plays Golf
	Outlook	Temperature	Humidity	Windy	
$\boldsymbol{x}_1$	Rainy	Cool	Normal	False	$y_1 = \text{Yes}$
$\boldsymbol{x}_2$	Overcast	Hot	High	True	$y_2 = \text{Yes}$

Final Decision tree



Which attribute to split?

- Gini Impurity of a set:

$$Gini(set) = 1 - \left(\frac{P}{T}\right)^2 - \left(\frac{N}{T}\right)^2$$

- Gini Impurity of a test:

$$Gini(test) = \sum_{all\ sets} Gini(set) \times \frac{no\ of\ elements\ in\ set}{total\ elements}$$

Which attribute to split?

- Disorder/Entropy of a binary set:

$$Entropy(set) = -\frac{P}{T}\log_2\frac{P}{T} - \frac{N}{T}\log_2\frac{N}{T}$$

- Disorder/Entropy of a test:

$$Entropy(test) = \sum_{all\ sets} Entropy(set) \times \frac{no\ of\ elements\ in\ set}{total\ elements}$$

- Information Gain: Entropy before split minus entropy after split.

Questions?